AUtoMLy | Medicaid Policy Copilot

INAI 6986 — Milestone 2: Model Testing, Validation and Refinement

J.O. • Dr. David M. Simmonds • Evidence snapshot captured 2026-10-16 15:03:32 CDT (shared with Progress Log 2; receipt digests in Appendix A)

PRE-TEST WORKING DRAFT — FINAL EVALUATION NOT RUN

The one-time final benchmark scoring has not been executed. The fresh benchmark is not yet created. Source/key and exposure approval, native matched-model proof and immutable protocol sealing remain open (Appendix A); genuinely new human output ratings follow generation. This document prepares the mandated structure without claiming a test score, threshold verdict or deployment readiness.

1. Summary of Results

No valid model-versus-base factual test result exists at this evidence cutoff. Sixteen completed nested-size training runs (four each for Muse, Gemma, Nemotron and Ornith) used fixed validation prompt IDs and zero test/control rows; they demonstrate token fit, not policy-answer accuracy. Fixed Nemotron N=1,600 checkpoint-400 and its matched untouched base have not been paired on a locked final benchmark; this candidate is not a demonstrated winner. The result against the planned ten-percentage-point factual-accuracy improvement and safety targets is therefore undetermined—not a pass or fail. The cross-family development table uses a machine-advisory probe (Section 5), not human or final accuracy; the completed Batch B human review is reported separately.

Decision measure	Current observed result	Precommitted target / meaning
Answerable factual accuracy; base versus selected model	Not measured on a shared locked final set	Model improvement at least +10 percentage points versus untouched base
Correct source locator	Not measured on a representative final set	At least 95% on the relevant scored denominator
Unsupported factual claims	Not measured on a representative final set	Fewer than 2% of scored claims
Correct abstention	Not measured; original 30 controls are all answerable	At least 90% on a meaningful unsupported-question stratum

Business translation is pending: report correct supported answers, unsafe claims and appropriate abstentions per 100 relevant cases only after observed denominators exist. Manual 30/30 on simple page/code lookups is an easy-task ceiling, not a same-split comparison or measured analyst time savings.

2. Experimental Setup

Target: a sourced answer to a public Alabama Medicaid policy/provider-billing question, or an abstention when the frozen source does not support one. It is not an autonomous eligibility, payment or clinical determination. A source file, page/section locator, answerability label and source quote define the adjudication record. The deep-learning family comprises locally adapted language models; ALPHIN is the model component, not the name of the overall project.

Pipeline and Protection

The canonical ordinary supervised corpus has 1,999 answerable records and a document-grouped historical 1,684/124/191 train/validation/test split. All 30 original student-authored control items map by source URL plus SHA-256 to historical test documents, with none mapped to clean train/validation; their wording and answer keys are independently authored but their source documents are not independent of the historical test group. The historical 191 have already been exposed as diagnostics and cannot now serve as an untouched final test. A new representative, source-verified control/test specification and immutable membership must be recorded before scoring.

Preparation → pinned base and response-only tokenizer/chat template → nested train subset (N=200/509/750/1,600) → fixed 124-row validation → per-run checkpoint selected by within-run validation loss → candidate freeze → exactly one model/base final evaluation. Preserve document identity, prompt-ID hashes, image/code revision, decoding settings, answer-key/source provenance and selected adapter digest. Training rows or test/control rows must not be introduced by an evaluation step.

Metrics and Business Asymmetry

Committed primary: human-adjudicated answerable factual accuracy. The reviewed harness instead names full-cohort source-context answer success with a machine-advisory judge and a planned human subset (15 answerable, 3 unsupported items; 36 outputs, 216 dimension choices). This is a protocol-reconciliation gate before inference: preserve the human primary on its actual rated denominator, report full-cohort advice separately, and seal any changed definition and reason. Visible machine advice makes that human review AI-assisted, not blind. Secondary: verifiable locator accuracy, unsupported-claim rate, correct abstention on an adequate unsupported stratum, safety routing, general-task regressions and latency. A false confident policy claim is costlier than an abstention that triggers analyst review. Teacher-forced next-token accuracy and loss are training diagnostics, not answer-level validation scores; no cross-tokenizer factual winner may be inferred from them. The current offline evidence-replay notebook recomputes the historical split, 16-run training summaries and development advisory metrics, but cannot reproduce a final test that has not occurred.

3. Iteration Narrative

The disciplined experiment loop changed training-set size while preserving the same validation prompt IDs. Every run has a recorded rationale, selected checkpoint and train/validation token metrics in run_log.csv and its separate cohort artifact. This is an engineering refinement arc, not a validated answer-level model selection.

Family	Nested sizes completed	N=1,600 final / best step	What the runs establish
Gemma 4	200, 509, 750, 1,600	600 / 400	Token-fit trend within Gemma; selected adapter is not a factual winner.
Muse Glimmer	200, 509, 750, 1,600	600 / 200	Token-fit trend within Muse; pre-amendment September runs excluded.
Nemotron 3.5 Lightning	200, 509, 750, 1,600	1,200 / 400	Corrected-code validation/early stopping; no protected rows.
Ornith 1.5	200, 509, 750, 1,600	1,200 / 400	Corrected-code validation/early stopping; no protected rows.

Failed or Non-Comparable Attempts

Early September Muse experiments were superseded by changed generator/split/training methodology; their metrics are retained as history and excluded from the current matrices. The first Nemotron N=200 attempt failed before training on an unsupported warmup argument. Pre-fix Nemotron N=200 and Ornith N=200/N=509 attempts failed selected-checkpoint finalization when a nested train evaluation cleared scheduled validation. A targeted callback fix was regression-tested, then all eight architecture runs were repeated from pinned bases in fresh directories; no failed attempt is presented as clean evidence.

Within-run checkpoint selection used validation loss, early-stopping patience two and minimum improvement 0.005 for the eight corrected architecture runs. All eight archives have 28 matching host-to-local file hashes. These checks verify training integrity. Since the October 1 cutoff, the Week 8 thirty-error human review was completed as the Batch B card review (revision 11, verified 2026-10-07: 30 cards, 180 dimension ratings) and one targeted development change was attempted and measured (below). A frozen factual candidate and a measured Week 9 stability check remain open.

Iterations Since October 1

October 5–7: human dispositions arrived for the sixteen duplicate-answer groups and the flagged source-column items (Batch A receipt, 27 responses; nine supplementary question/answer pairs labelled AI-assisted/student-adopted, 8 identical to local-LLM drafts and 1 edited). The Batch B thirty-error card review (30 AI-triage-selected cards, not 30 confirmed errors) closed at revision 11 with 180 ratings; a correction round is recorded in the durable history (final revision 11). A targeted prompt-refinement experiment paired eight question sets across base and adapter (16/16 outputs complete): lexical diagnostics moved (refined base: eight outputs carrying an explicit cannot-verify phrase), machine semantic scoring completed, human_scores_created=false and model_winner_selected=false, so nothing was promoted. October 14: a final evaluation harness passed SPEC and QUALITY reviews (runtime code digest b4c06b7f5a85770a…), its protocol pre-declaring the robustness shift and scorer. October 16: authorized new-epoch recovery and 09:14 real-installation integration retired 225 old keys, refused 225 old admissions and preserved 226 copies. Source-only preflights authenticated eight originals/28 physical pages twice, with zero model/network calls. The earlier blocked transition was superseded, not a pending user decision; this recovery is not benchmark construction or native inference.

4. Test Results — Evaluation Not Yet Run

No accepted fresh benchmark membership or final outputs exist; the final event remains unexecuted. A legitimate same-split table requires one preselected candidate, the untouched base, the same fixed test records and decoding settings, a signed answer key and a committed scorer. Do not fill missing measurements with token loss, a single unanswerable example, or the historical 191 diagnostics.

Same locked final test condition	Answerable factual accuracy	Correct locator	Unsafe claim / abstention
Untouched base; fixed generation settings	Not run	Not run	Not run
Fixed Nemotron N1,600 checkpoint-400	Not run	Not run	Not run
Business thresholds (predeclared)	At least +10 pp versus base	At least 95%	Unsafe claims <2%; correct abstention ≥90%

Confusion-Matrix Equivalent

After the one-time run, publish counts and denominators for answerable cases answered correctly with supported locator; answerable cases answered wrongly or without support; answerable cases unnecessarily abstained; and unanswerable cases correctly abstained versus answered unsafely. Maintain separate answerability strata so a pooled accuracy cannot hide an abstention failure. No cell currently has a measured count; blank is not zero.

Business Units and Test Discipline

Once measured, compare base/model deltas in correct supported answers per 100 answerable cases and unsafe responses per 100 unsupported cases, then test the precommitted pass/fail gates. No time-saving percentage is supported by the retrospective manual lookup proxy. If the first test reveals a defect, preserve and diagnose it; do not tune on those final outcomes or choose a different winner after exposure. Record an honest negative result if the threshold is missed.

5. Error and Robustness Analysis

The Week 8 thirty-error human review was completed on 2026-10-07 as the Batch B card review: 30 AI-triage-selected review cards (not 30 confirmed errors — some cover preserved or correct material) with 6 dimension ratings each, 180 ratings total, durably verified at revision 11. Named failure modes with counts currently exist as machine-advisory labels from the 2026-10-08 rescoring of the 192-slot development probe: across 150 accepted advisory judgments the error mix is wrong value 57, abstention issue 32, hallucinated rule 26, no error 26, invalid locator 8, missing evidence 1. These labels were produced by a local advisory judge, not by human adjudication, and they are development-diagnostic, not final-test counts. The sixteen duplicate-answer groups (57 generated records) are data-quality flags, now human-disposed in the Batch A receipt; the recorded H2011 ruling is to verify the table reading and correct Daily Max to 4 in a new version.

Named failure mode (machine-advisory label count of 150)	What it suggests — modeling, data or scope change
Wrong value (57 of 150)	Inspect source version/effective date and competing rule; correct data or tighten the answer scope rule.
Abstention issue (32 of 150)	Human-adjudicate answerable versus unanswerable strata separately; revise the decision boundary only on development data.
Hallucinated rule (26 of 150)	Check citation against the exact passage; enforce source-grounding and escalate unsupported confident claims.
Invalid locator (8 of 150)	Verify the generated locator resolves, not merely that a URL is present; repair parsing or retrieval.
Missing evidence (1 of 150) and No error (26 of 150)	Treat missing-evidence cases as extraction/citation gaps; No Error is only an advisory category, not the stricter supported-success measure (17/192).

Robustness Protocol — Predeclared, Not Yet Measured

The reviewed harness specifies a frozen reduced-context perturbation: retain the first 60% of source characters, rounding down; score evidence-surviving and evidence-erased cases separately. It does not specify paraphrase or a later revision as interchangeable completed tests. Exact subset membership and evidence-survival bindings are not yet sealed. No perturbation has been evaluated and no delta exists. A future full-cohort machine-advisory change is not a human-primary robustness result: the same declared metric and actual rated denominator must be compared, or the human-primary robustness gap stays open. Unit tests and more cannot-verify phrases do not satisfy this requirement; evaluation does not reopen tuning.

6. Subgroup Performance and Risk

Policy-support risks differ by answerability, document family, exception/date requirements and source vintage. Predefine those segments on the protected benchmark; report per-segment counts, accuracy, unsupported claims, locator correctness and abstention with denominator and uncertainty. A single pooled token-level score does not establish equitable or reliable policy support. There are no verified model-output subgroup gaps to report at this cutoff.

Segment / expected use	Measurement after freeze	Decision if gap appears
Answerable versus plausible unanswerable	Separate factual accuracy and correct-abstention denominators	Block unsupported confident answers; route for analyst review.
Provider manuals versus Administrative Code	Answer/locator accuracy and error counts by document family	Inspect extraction, revision and source coverage.
Simple lookup versus exception/date/multi-rule reasoning	Stratified accuracy and unsupported-claim rate	Limit scope of model-aided drafting until reasoning cases pass.
Current versus later source revisions	Stale-source and date-mismatch failure count	Versioned retrieval and manual approval; retrain only after review.

Risk Framework Applied to This Decision

The National Institute of Standards and Technology AI Risk Management Framework (NIST AI RMF 1.0) offers Govern, Map, Measure and Manage functions. For this use, Govern requires an accountable analyst and source-version owner; Map identifies incorrect policy advice and unsupported claims as harms; Measure requires the answerability and source-family slices above; Manage blocks deployment when factual or abstention gates fail and triggers monitoring/escalation. This is a risk-management mapping, not a claim that the system has passed an audit or meets a specific regulatory classification. Source: https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf

7. Limitations and Deployment Readiness

No completed factual comparison, final-test score or measured robustness/subgroup result supports deployment. Current public source documents can change; a static adapter is not a current-law source. The original thirty control questions are personally authored and source-verified but are simple and concentrated, and their source documents are historically test-mapped. Fresh official-source acquisition and authoring tools have since been prepared under the recorded 2026-10-14 decision: the approved new items will be local-AI-generated and source-verified — explicitly not independently student-authored — and no additional student-authored questions were requested. Originals and all human submissions are preserved with their distinct authorship labels.

Completed: Batch A human dispositions, Batch B revision 11, one negative development refinement, reviewed harness code and the authorized October 16 recovery integration. Before inference: create and independently source-verify the representative benchmark, resolve flagged source exposure, prove native candidate/base parity, reconcile the human-primary versus machine-advisory protocol, and seal exact settings/membership/scoring/robustness. After the one-time run: obtain genuinely new scoped human output ratings, compute final results and execute their analysis notebook. Current review and replay do not substitute for any of those measurements. Autonomous policy decisions remain out of scope even if a later drafting pilot passes.

Operating and Monitoring Conditions

Any future pilot must use versioned authoritative retrieval, clear source locators, analyst approval before use, abstention/escalation for unsupported or conflicting guidance, access controls and an audit trail. Monitor factual/locator/unsupported-claim/abstention rates by source family, answerability and date; investigate source-version drift and new document families. Reconsider deployment or retraining if a precommitted safety threshold is missed, source revisions invalidate citations, or subgroup gaps appear. These are proposed safeguards, not implemented or measured runtime capabilities.

8. Repository and Reproducibility

Project repository: https://github.com/porespellar/INAI6986_project_deliverables. Hosting metadata confirms the source repository is private and the J.O. deliverables repository is public. Current evidence replay: https://github.com/porespellar/INAI6986_project_deliverables/blob/main/notebook/current_evidence_replay_20261016.ipynb . It recomputes current training summaries and development advice from fingerprint-verified per-epoch/per-output records, without rerunning inference. It is not a final-test notebook. No final result exists to reproduce. Private instructor access and actual Canvas submission remain separate, unverified gates.

AI Use Disclosure

Hermes Agent (GPT-6 Luna draft; GPT-6.1 Sol review) and local Qwen assisted writing, code and evidence checks; local workers generated development data and advice. I designed and directed the project, authored and source-checked the original controls, supplied human ratings and retain responsibility. AI did not supply human ratings or final-test results; I must approve this draft before submission.

Appendix A — Evidence and Outstanding Acceptance Gates

All dated claims below read one shared machine-generated snapshot (artifacts/milestone2_completion/class_package_20261016/evidence_snapshot_20261016.json, captured 2026-10-16 15:03:32 CDT, sha256 f0c66d60b669e1d2…), which stores the sha256 of every underlying receipt.

Claim or gate	Authoritative evidence / present disposition
Frozen source and historical split	approved_candidates.jsonl (1,999 answerable records); 1,684 train / 124 validation / 191 already-exposed historical test; document-grouped split.
Original independent set	student-authored-control-set-2026-09-22.jsonl: 30 personally authored and source-verified lookup questions, preserved unchanged. Authorized new fresh-source items will be local-AI-generated/to be source-verified under the October 14 decision; the cohort is not yet created and is never independent student authorship.
Training evidence	results.jsonl: 28 Gemma/Muse epoch records; architecture-clean-run-matrix.json: eight Nemotron/Ornith runs, 24 epochs; run_log.csv: 16 experiments; zero test/control rows used.
Failed attempts	docs/engineering/records/superseded-pre-fix-architecture-runs-20261001.json; superseded and excluded from current matrices.
Leakage disposition — closed	Batch A receipt received 2026-10-05: 27 responses incl. all 16 duplicate-group dispositions; 9 supplementary AI-assisted/student-adopted pairs; one historical pending flag retained; Verify table reading; correct to Daily Max = 4 in a new version (recorded in the feedback response ledger, not re-opened here)
Thirty-error human review — closed	Batch B rev 11 verified 2026-10-07: 30 cards, 180 ratings; submission digest cd5ab8062c661994…; final_evaluation_authorized_by_this_submission=false.
Development advisory probe — machine only	192 slots / 150 accepted judgments / 19 budget + 23 parser failures / 17 fully supported / 48 abstentions; judge qwen3.8-flash-next-nvfp4; human_scored=false; winner=null. Advisory only — never a final or human metric.
Targeted refinement — measured, not promoted	8 paired question sets, 16/16 outputs complete (2026-10-07); machine semantic scoring complete; human_scores_created=false; no model winner selected.
Final harness — reviewed, unexecuted	SPEC and QUALITY reviews 2026-10-14; runtime code digest b4c06b7f5a85770a…; final run executed=false.
Recovery integration — 2026-10-16	225 prior run keys retired, 225 old admissions refused, 226 copies preserved, 0 model/network calls; new-epoch provisioning and real-installation integration passed at 09:14 CDT; earlier blocked resume is superseded. Source-only checks made no model calls; no final cohort was created.
Exposure/source gate — OPEN	12 fresh-source documents audited (2026-10-14): 11 with no project-relative exposure evidence; ctx-admcode-ch16-pharmacy-20260826 flagged EXPOSURE_EVIDENCE_REVIEW_REQUIRED; final source approval=false.
Final test, notebook and Canvas	One-time final test unused and unexecuted; current development raw-to-metric replay is available, but no final-result notebook can exist before that run. Canvas deadlines not re-verified at this snapshot (recorded response: cannot check right now). Due October 25, 2026, 11:59 PM CT per the assignment record.

Gate that must pass before any final inference	Present state
Create representative local-AI benchmark and independently verify exact source/keys, including the flagged pharmacy source exposure; source approval remains false	Open — cohort not created; source approval false
Prove native runtime admission and exact matched Nemotron checkpoint-400/untouched-base assets and parity	Open — frozen runtime admission not yet exercised
Seal exact accepted membership, keys, prompt, generation settings, scoring, human sample and robustness perturbation before the one-time final run	Open

This appendix records exact evidence boundaries. It neither changes the held-out data nor turns a pre-test working draft into a completed Milestone 2 submission.

Appendix B — Milestone 1 Feedback Response Crosswalk

This concise crosswalk reconciles all nine feedback items with verified receipts; the full ledger and receipt identities remain in the private source repository. Completed work and unresolved criteria are kept distinct.

ID	Reviewer Concern	Verified Response And Remaining Limitation
M1-01	Lookup-only questions	Preserve all 30 originals; separately authorized reasoning/date/exception/conflict benchmark is not yet created. Local-AI authorship is disclosed; no instructor waiver.
M1-02	Unsupported questions	Nine historical supplements received: eight adopted AI drafts, one edited. They are not independent student authorship or a verified fresh unsupported stratum.
M1-03	Narrow source coverage	All 30 controls map by URL and hash to historical-test sources; 21/30 on three pages. Wording independence does not imply unseen sources.
M1-04	Ceiling baseline / URL formatting	Manual 30/30 is an easy lookup reference; 84 seconds is a retrospective proxy. Current 192-slot advisory results are not final/human accuracy or time savings.
M1-05	Duplicate-answer leakage	All 16 groups / 57 records have Batch A human dispositions. Application of versioned remedies to canonical data and selected weights is not evidenced.
M1-06	Insufficient alternatives	Four genuine families at four nested sizes; 16 completed runs. Three adapters tie at 4/24 advisory support. Fixed Nemotron400 is a governed candidate, not a winner.
M1-07	Overlong control appendix	Two compact source-split tables preserve original wording and show coverage; original detailed evidence remains archived.
M1-08	Source-column and exposure defects	Near-duplicate ruling and H2011 Daily Max=4 correction direction received. Do not claim existing weights were repaired. Historical test remains exposed.
M1-09	Priority before next milestone	Completed review/refinement work is preserved. Source-only recovery integration passed; fresh benchmark, native parity, seal, final outputs and new human ratings remain missing.

Appendix C — Rubric Evidence And Credit Rationale

This crosswalk requests credit only for work evidenced here. A criterion with an open gate is identified rather than treated as satisfied; the appendix is not an instruction to the grader or a score guarantee.

M2 criterion	Evidence in this report	Defensible claim / remaining limitation
Evaluation rigor (25)	Sections 2, 4, 8; Appendix A snapshot and receipt digests.	Test-set protection, single-evaluation discipline and a predeclared protocol are documented and reproducible from the repo; the benchmark itself and final evaluation are missing. Predeclared harness metrics still need reconciliation with the human primary; protection controls are not measured results.
Interpretation & business framing (20)	Section 1 decision table; Section 4 business-unit translation plan.	Baseline/threshold comparison structure and business asymmetry are stated plainly; verdict is honestly undetermined — no measured result exists to translate yet.
Iteration & refinement (20)	Section 3 run-log arc, failed attempts, and the October 5–16 iteration record.	Failed and superseded experiments are retained with what they ruled out; refinement and harness iterations since October 1 are evidenced by receipts.
Error, robustness & subgroup analysis (20)	Section 5 (Batch B rev 11; advisory-labeled counts), Section 6 segments, NIST AI RMF mapping.	A completed 30-card human review and countable machine-advisory failure labels exist as distinct evidence; robustness is predeclared but unmeasured and final subgroup gaps have no measured values yet. This does not fulfill the actual robustness requirement.
Limitations & readiness (10)	Sections 6–7 operating conditions and retraining triggers.	Validity limits and monitoring plan are concrete; deployment readiness is explicitly withheld pending the final evaluation.
Communication & format (5)	Document structure, AI Use Disclosure, appendices.	Disclosure names tools, uses and human verification; organization follows the mandated template with appendices beyond the 7-page narrative.